Year 2 Pure Mathematics

Chapter 7

Trigonometry and modelling

Practice Booklet

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7.1 Addition formulae

Practice Book, printed page 49.

1.

a. By substituting $x = 90° - A$ and $y = B$ into the expansion of $\sin(x + y)$, show that $\cos(A - B) = \cos A \cos B + \sin A \sin B$

b. By substituting $x = A$ and $y = -B$ into the expansion of $\cos(x - y)$, show that $\cos(A + B) = \cos A \cos B - \sin A \sin B$

Hint

$\sin(90° - x) = \cos x$ and $\cos(90° - x) = \sin x$
$\cos(-y) = \cos y$ and $\sin(-y) = -\sin y$
← Year 1, Section 9.6

2.

a. Write $\sin\left(x - \frac{\pi}{3}\right)$ in the form $a\sin x + b\cos x$, where a and b are constants to be found.

b. Write $\cos\left(x - \frac{\pi}{6}\right)$ in the form $p\cos x + q\sin x$, where p and q are constants to be found.

c. Write $\tan\left(x + \frac{\pi}{3}\right)$ in the form $\frac{\tan x + m}{1 - m\tan x}$, where m is a constant to be found.

Hint

The addition formulae for sin, cos and tan are given in the formulae booklet:
$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

3. Express each of the following as a single trigonometric function:

a. $\frac{\tan 7\theta - \tan 3\theta}{1 + \tan 7\theta \tan 3\theta}$

b. $\cos\left(\frac{3a + 5b}{3}\right)\cos\left(\frac{3a - 5b}{3}\right) - \sin\left(\frac{3a + 5b}{3}\right)\sin\left(\frac{3a - 5b}{3}\right)$

c. $\frac{\sqrt{2}}{2}\sin x - \frac{\sqrt{2}}{2}\cos x$

Hint

Match each expansion to the correct addition formula and write the resulting expression as simply as possible.

4. E/P Express $\dfrac{\cos(x-y)}{\sin(x+y)}$ in terms of $\tan x$ and $\tan y$. (3 marks)

5. P Use the addition formulae to write:

a. $\dfrac{1}{2}(\cos x - \sqrt{3}\sin x)$ in the form $\cos(x \pm \theta)$, where $0 < \theta < \dfrac{\pi}{2}$

b. $\dfrac{1}{2}(\sqrt{3}\sin x - \cos x)$ in the form $\sin(x \pm \theta)$, where $0 < \theta < \dfrac{\pi}{2}$

6. E/P Given that $\sin\left(x + \dfrac{\pi}{6}\right) = \cos x$, show that $\tan x = \dfrac{1}{\sqrt{3}}$ (3 marks)

7. E/P Show that $2\sin\left(x + \dfrac{5\pi}{6}\right) + 2\sin\left(x + \dfrac{10\pi}{6}\right) = (1-\sqrt{3})\sin x + (1-\sqrt{3})\cos x$. (4 marks)

7.2 Using the angle addition formulae

Practice Book, printed page 49.

1. Without using your calculator find the exact value of:

a. $\sin\dfrac{\pi}{12}\cos\dfrac{\pi}{4} + \cos\dfrac{\pi}{12}\sin\dfrac{\pi}{4}$

b. $\cos 96^\circ \cos 51^\circ + \sin 96^\circ \sin 51^\circ$

c. $\dfrac{\tan\frac{\pi}{18} + \tan\frac{\pi}{9}}{1 - \tan\frac{\pi}{18}\tan\frac{\pi}{9}}$

Hint

Use the addition formulae in reverse and your knowledge of exact values of trigonometric ratios.

2.

a. Use the formula for $\sin(A+B)$ to show that $\sin 135^\circ = \dfrac{\sqrt{2}}{2}$

Hint

For part a, write 135° as $90^\circ + 45^\circ$.

b. Use the formula for $\cos(A-B)$ to show that $\cos 15^\circ = \dfrac{\sqrt{6}+\sqrt{2}}{4}$

c. Use the formula for $\tan(A+B)$ to show that $\tan 75^\circ = 2 + \sqrt{3}$

3. Given that $\cos A = \dfrac{3}{5}$ and $\sin B = \dfrac{8}{17}$, where A and B are both acute angles, calculate the exact value of:

a. $\cos(A+B)$

b. $\tan(A-B)$

c. $\operatorname{cosec}(A+B)$

Hint

You can find the values of $\sin A$ and $\cos B$ using $\sin^2 x + \cos^2 x \equiv 1$.

4. P Given that $\cos A = \dfrac{1}{2}$, and that A is acute, find the exact value of:

a. $\cos\left(\dfrac{\pi}{6} - A\right)$

b. $\tan\left(\dfrac{\pi}{3} + A\right)$

c. $\sin\left(\dfrac{\pi}{2} - A\right)$

5. P Given that $\sin A = \frac{24}{25}$, where A is obtuse and $\cos B = -\frac{5}{13}$, where B is reflex, calculate the exact value of:

a. $\sin(A + B)$

b. $\tan(A - B)$

c. $\sec(A + B)$

6. E/P

a. Express $\tan\left(\frac{\pi}{4} - \frac{\pi}{6}\right)$ in terms of $\tan\frac{\pi}{4}$ and $\tan\frac{\pi}{6}$ (2 marks)

b. Hence show that $\tan\frac{\pi}{12} = 2 - \sqrt{3}$. (2 marks)

7. E/P

a. Use $\sin(x + \alpha) \equiv \sin x \cos\alpha + \cos x \sin\alpha$ to show that $\sin 105° = \frac{\sqrt{6} + \sqrt{2}}{4}$ (4 marks)

b. Hence, or otherwise, show that $\operatorname{cosec} 105° = \sqrt{6} - \sqrt{2}$ (3 marks)

7.3 Double-angle formulae

Practice Book, printed page 50.

1. Write each of the following expressions as a single trigonometric ratio.

a. $2\sin\frac{\pi}{3}\cos\frac{\pi}{3}$

b. $4\cos^2 7° - 2$

c. $\frac{1 - \tan^2 13.5°}{2\tan 13.5°}$

Hint

$\sin 2A \equiv 2\sin A\cos A$
$\cos 2A \equiv \cos^2 A - \sin^2 A \equiv 2\cos^2 A - 1 \equiv 1 - 2\sin^2 A$
$\tan 2A \equiv \frac{2\tan A}{1 - \tan^2 A}$

2. Write each of the following in terms of a single trigonometric function.

a. $6\sin 5\theta\cos 5\theta$

b. $\tan\theta(1 + \cos 2\theta)$

c. $\frac{\sin 2\theta}{1 - \cos 2\theta}$

3. Given that $\sin A = \frac{5}{13}$, and that A is obtuse, find the exact value of:

a. $\tan 2A$

b. $\cos 2A$

c. $\operatorname{cosec} 2A$

Hint

A is in the second quadrant, so only $\sin A$ will be positive.

4. P Eliminate θ from the following pairs of equations and express y in terms of x.

a. $x = 5\sin\theta$, $y = 2 - 10\cos 2\theta$

b. $x = \cos 2\theta$, $y = 4\sec\theta$

5. E/P

a. Show that $\cos^4 x - \sin^4 x \equiv \cos 2x$. (2 marks)

b. Hence find the exact value of $\cos^4\frac{\pi}{12} - \sin^4\frac{\pi}{12}$ (2 marks)

6. E/P Given that $a = 2\sin x$, $b = 2\cos x$, $c = \cos 2x$ and $d = \tan 2x$,

a. express c in terms of a (2 marks)

b. show that $d = \dfrac{2ab}{b^2 - a^2}$ (3 marks)

7. E/P Given that $\pi < \theta < \dfrac{3\pi}{2}$, find the value of $\sin\dfrac{\theta}{2}$ when $\cos\theta = \dfrac{24}{25}$ (4 marks)

7.4 Solving trigonometric equations

Practice Book, printed page 51.

1. Solve in the interval $0 \leq x \leq 2\pi$, giving your answers to 1 decimal place:

a. $\sin\left(x + \dfrac{\pi}{3}\right) + 4\sin x = 0$

b. $\cos\left(\dfrac{\pi}{6} + x\right) = \sin x$

c. $\cos\left(x + \dfrac{\pi}{4}\right) + \sin\left(x + \dfrac{\pi}{4}\right) = \dfrac{1}{2}$

Hint

Use the addition formulae to expand and simplify the left-hand side of each equation. Then write an equation in $\tan x$ and solve.

2. Solve in the interval $0 \leq \theta \leq 180°$, giving your answers to 1 decimal place where necessary:

a. $1 + \cos 2\theta = 7\cos^2\theta - 1$

b. $\sin 2\theta = \dfrac{1}{4}\sin\theta$

c. $\tan\theta \tan 2\theta = 3$

Hint

Start by using the double-angle formulae to write the terms involving 2θ in terms of θ.

3.

a. Show that $\cos\left(x - \dfrac{\pi}{6}\right) \equiv \dfrac{1}{2}(\sqrt{3}\cos x + \sin x)$.

b. Hence solve the equation $\sqrt{3}\cos x + \sin x = 1$ in the interval $0 \leq x \leq 2\pi$, giving your answers in terms of π.

Hint

For part b, solve $\cos\left(x - \dfrac{\pi}{6}\right) = \dfrac{1}{2}$

4. E/P

a. Show that $\sin(x - 45°) = \cos x$ can be written as $\tan x = a + \sqrt{b}$, where a and b are real constants to be found. (3 marks)

b. Hence, or otherwise, solve $\sin(x - 45°) = \cos x$ in the interval $0 \leq x \leq 360°$, giving your answers to 1 decimal place. (3 marks)

5. E/P

a. Show that $2\sin\left(x - \dfrac{\pi}{6}\right) = \sin\left(x + \dfrac{\pi}{2}\right)$ can be written in the form $\sqrt{3}\sin x - 2\cos x = 0$. (3 marks)

b. Hence solve $2\sin\left(x - \dfrac{\pi}{6}\right) = \sin\left(x + \dfrac{\pi}{2}\right)$ in the interval $-\pi \leq x \leq \pi$. Give your answers to 2 decimal places. (3 marks)

6. E/P

a. Given that $\tan 2x - \tan x = 0$, show that $\tan x + \tan^3 x = 0$. (3 marks)

b. Hence solve $\tan 2x = \tan x$ in the interval $0 \leq x \leq 2\pi$. Give your answers in terms of π. (3 marks)

7. E/P

a. Express $2\sin^2 x - \cos 2x - \cos x = 0$ in the form $a\cos^2 x + \cos x - b = 0$. (3 marks)

b. Hence solve the equation $2\sin^2 x - \cos 2x - \cos x = 0$ in the interval $-\pi \leq x \leq \pi$. Give your answers to 2 decimal places when they are not exact. (4 marks)

8. E/P Solve the equation $\cos^2 x - 2\sin^2 x - \sin 2x = 1$ in the interval $-\pi \leq x < \pi$. Give your answers to 2 decimal places when they are not exact. (6 marks)

7.5 Simplifying $a\cos x \pm b\sin x$

Practice Book, printed page 52. **Unless otherwise stated, give all angles to 1 decimal place and write non-integer values of R in surd form.**

1. Find the value of R, $R > 0$, and the value of $\tan\alpha$, given that:

a. $20\sin\theta + 21\cos\theta \equiv R\sin(\theta + \alpha)$

b. $\sqrt{3}\cos\theta - \sqrt{5}\sin\theta \equiv R\cos(\theta + \alpha)$

c. $2\sin\theta - 3\cos\theta \equiv R\sin(\theta - \alpha)$

Hint

Expand the right-hand side and equate the coefficients of $\sin\theta$ and $\cos\theta$ to obtain two equations in R and α. Divide the equations to eliminate R and find the value of $\tan\alpha$.
Square and add the equations to eliminate α and find the value of R.

2. Find the value of R, $R > 0$, and α, where $0 < \alpha < 90°$, given that:

a. $2\sin\theta + \cos\theta \equiv R\cos(\theta - \alpha)$

b. $12\cos\theta - 5\sin\theta \equiv R\cos(\theta + \alpha)$

c. $4\sin\theta - 2\cos\theta \equiv R\sin(\theta - \alpha)$

Hint

Use the $\tan^{-1}$ function on your calculator to find the value of α in each case.

3.

a.

i. Express $\sqrt{3}\sin\theta - \cos\theta$ in the form $R\sin(\theta - \alpha)$, where $R > 0$, $0 < \alpha < \frac{\pi}{2}$

ii. Hence sketch the graph of $y = \sqrt{3}\sin\theta - \cos\theta$ in the interval $0 \leq \theta \leq 2\pi$.

Hint

The graph of $y = R\sin(\theta - \alpha)$ is the graph of $y = \sin x$ stretched by scale factor R in the vertical direction, and translated by the vector $\begin{pmatrix}\alpha\\0\end{pmatrix}$.

b.

i. Express $2\cos\theta - \sqrt{12}\sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$, $0 < \alpha < \frac{\pi}{2}$

ii. Hence sketch the graph of $y = 2\cos\theta - \sqrt{12}\sin\theta$ in the interval $0 \leq \theta \leq 2\pi$.

c.

i. Express $2\sin\theta + 2\cos\theta$ in the form $R\sin(\theta + \alpha)$, where $R > 0$, $0 < \alpha < \frac{\pi}{2}$

ii. Hence sketch the graph of $y = 2\sin\theta + 2\cos\theta$ in the interval $0 \leq \theta \leq 2\pi$.

4. E/P $\mathrm{h}(\theta) = 2\sin\theta + 4\cos\theta$

a. Given that $\mathrm{h}(\theta) = R\sin(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < 90°$, find the exact value of R and the value of α to 2 decimal places. (3 marks)

b. Hence solve $2\sin\theta + 4\cos\theta = 3$, for $0 \leq \theta \leq 360°$, giving your answers to 1 decimal place. (3 marks)

5. E/P

a. Express $3\cos 2\theta + 2\sin 2\theta$ in the form $R\cos(2\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$, giving the value of α to 3 decimal places. (3 marks)

b. Hence solve $3\cos 2\theta + 2\sin 2\theta = 1$, for $-\pi \leq \theta \leq \pi$, giving your answers to 2 decimal places. (5 marks)

c. Given that k is a constant and the equation $3\cos 2\theta + 2\sin 2\theta = k$ has no solutions, state the range of possible values of k. (2 marks)

6. E/P

a. Express $8\sin\theta - 15\cos\theta$ in the form $R\sin(\theta - \alpha)$, where R and α are constants, $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give the value of α to 2 decimal places. (3 marks)

$$\mathrm{g}(\theta) = \frac{10}{23 + 8\sin\theta - 15\cos\theta}$$

b.

i. Find the maximum value of $\mathrm{g}(\theta)$.

ii. Find the value of θ, in the interval $0 \leq \theta \leq 2\pi$, at which this maximum value occurs, giving your answer to 2 decimal places.

(3 marks)

c.

i. Find the minimum value of $\mathrm{g}(\theta)$.

ii. Find the value of θ, for $0 \leq \theta \leq 2\pi$, at which this minimum value occurs, giving your answer to 2 decimal places.

(3 marks)

7. E/P

a. Show that that $2\sin\left(x - \frac{\pi}{6}\right) - (\sqrt{3} - 2)\sin x$ can be written as $2\sin x - \cos x$. (3 marks)

b. Solve $2\sin\left(x - \frac{\pi}{6}\right) - (\sqrt{3} - 2)\sin x = \frac{1}{2}$ in the interval $-\pi \leq x \leq 2\pi$. Give your answers to 2 decimal places. (4 marks)

c. Find the maximum value of $\left(2\sin\left(x - \frac{\pi}{6}\right) - (\sqrt{3} - 2)\sin x\right)^4$ (2 marks)

7.6 Proving trigonometric identities

Practice Book, printed page 53.

1. Prove the following identities.

a. $\dfrac{\sin 2x - \tan x}{\tan x} \equiv \cos 2x$

b. $\dfrac{\sin 6\theta}{1 - \cos 6\theta} \equiv \cot 3\theta$

c. $\dfrac{1}{4}\sec x \tan 2x \equiv \dfrac{1}{2}\sec 2x \sin x$

Hint

Use the double-angle formulae.
For part b, write $A = 3\theta$.
For part c, write the left-hand side in terms of $\sin x$ and $\cos x$. Dividing the numerator and denominator by a common term might be helpful.

2. Prove the following identities.

a. $\cos(x - 30°) - \cos(x + 30°) \equiv \sin x$

b. $\dfrac{\sin(a + b)}{\sin(a - b)} \equiv \dfrac{\cot b + \cot a}{\cot b - \cot a}$

c. $\cot\left(\frac{\pi}{4}-\theta\right) \equiv \frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}$

Hint

Use the addition formulae.

3. Prove the following identities.

a. $2\sin\left(3\theta+\frac{\pi}{3}\right) \equiv \sin 3\theta + \sqrt{3}\cos 3\theta$

b. $\cos 3A \equiv 4\cos^3 A - 3\cos A$

Hint

For part b, write $\cos 3A$ as $\cos(2A+A)$.

c. $\cot 2x + \tan x \equiv \operatorname{cosec} 2x$

4. E/P Prove that $\sqrt{3}\cos x - \sin x \equiv 2\cos(x+30°)$. (4 marks)

5. E/P

a. Prove that $\frac{\sin 2x}{1+\cos 2x} \equiv \tan x$. (3 marks)

b. Hence find the value of $\frac{\sin\frac{2\pi}{3}}{1+\cos\frac{2\pi}{3}}$ (2 marks)

6. E/P Prove that $2\sin\left(2\theta-\frac{\pi}{4}\right) \equiv 2\sqrt{2}\sin\theta\cos\theta - \sqrt{2} + 2\sqrt{2}\sin^2\theta$. (4 marks)

7. E/P

a. Prove that $\cos^4 x - \sin^4 x \equiv \cos 2x$. (3 marks)

b. Hence, or otherwise, given that x is acute and $\tan x = \frac{1}{\sqrt{3}}$, find the value of $\cos 2x$. (2 marks)

7.7 Modelling with trigonometric functions

Practice Book, printed page 54.

1. Georgia performs somersaults in the air while trampolining. Georgia's height above the trampoline, h metres, at time t seconds after she begins performing somersaults can be modelled by the equation

$$h = 0.3 - 0.3\cos(240t)°$$

a. State the maximum height Georgia reaches above the trampoline.

Hint

For part a, $\cos\theta$ has a maximum value of 1 and a minimum value of -1. So the maximum value of h occurs when $\cos(240t)° = -1$.

b. State the time it takes Georgia to reach this maximum height for the first time.

c. Given that Georgia performs one somersault every time she jumps, find according to the model, how many somersaults she will complete in 30 seconds.

Hint

For part c, start by using the period of curve to find out how long it takes to complete one somersault.

d. Criticise the model with respect to Georgia's path through the air.

2. The depth of water beneath a toy boat floating in a stream can be modelled by the equation

$$d = 40.5 + 2.5\sin(\pi t)$$

where d cm is the depth of the water and t is time in seconds.

a. Calculate the depth of water beneath the boat after 3 seconds.

b. Find the maximum depth of water beneath the boat according to this model.

Hint

For part b, use the maximum value of the sine function to find the maximum depth of water.

c. Calculate the times that the depth of water reaches this maximum in the first 5 seconds.

d. A larger toy boat needs a minimum depth of 40 cm of water to be able to float properly. State whether the water in the stream is deep enough.

3. Charlotte's blood pressure can be modelled by the function

$$P = 105 + 25\sin(450t)^\circ$$

where P is blood pressure in mm Hg and t is time in seconds.

a. Write down Charlotte's maximum blood pressure. This is known as systolic pressure.

b. Write down Charlotte's minimum blood pressure. This is known as diastolic pressure.

c. Calculate Charlotte's blood pressure after 4 seconds.

d. Find the times that Charlotte's blood pressure is 110 mm Hg during the first 2 seconds. Give your answers in seconds to 2 decimal places.

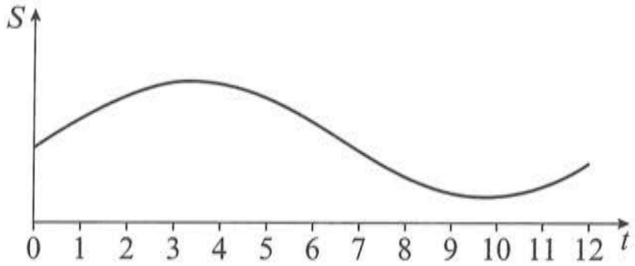
Hint

There are six different times in part d.

4. E/P A toy company records the average daily sales of a new toy during its first year of production. It models the data with the equation

$$S = 6 - 3\sin(0.5t + 3), \quad 0 \leq t \leq 12$$

where S is the number of toys sold in thousands, and t is time in months after 1 January 2018. The graph shows sales over the year.



a. Write down the maximum number of recorded sales, and state in which month this occurred. (2 marks)

The employees of the company are paid a bonus in any month during which the daily sales average exceeds 8500 on any day.

b. State the months in which a bonus was paid. (4 marks)

c. Calculate the average daily sales on the final day in December. (2 marks)

d. Tom says that the model predicts that sales will increase during the first few months of 2019. State whether you think he is correct. (1 mark)

5. E/P

a. Express $\frac{5\sqrt{2}}{2}\cos\theta - \frac{5\sqrt{2}}{2}\sin\theta$ in the form $R\cos(\theta+\alpha)^\circ$, where $R > 0$ and $0 < \alpha < 90^\circ$. (4 marks)

b.

i. Find the minimum value of $\frac{5\sqrt{2}}{2}\cos\theta - \frac{5\sqrt{2}}{2}\sin\theta$.

ii. Find the value of θ for $0 < \theta < 360^\circ$ at which the minimum occurs.

(3 marks)

The height of water at the entrance to a harbour over a period of 24 hours can be modelled by the equation

$$h = 9 - \frac{5\sqrt{2}}{2}\cos(30t) + \frac{5\sqrt{2}}{2}\sin(30t)$$

where h, metres, is the height of the water and t is the number of hours after midnight.

c. Write down the maximum height of water over the 24 hours, and the first time that this occurs. (3 marks)

d. Write down the minimum height of water over the 24 hours, and the final time this occurs. (3 marks)

e. Find:

i. the first time after which the water rises above 9 metres

ii. the first time after which the water falls below 9 metres.

(4 marks)

6. E/P

a. Express $0.2\cos x + 0.35\sin x$ in the form $R\cos(x-\alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give α to 1 decimal place. (4 marks)

The temperature of an electric heater can be modelled by the equation

$$T = 30 + 0.2\cos 2m + 0.35\sin 2m$$

where T is the temperature in Celsius and m is the time in minutes after the heater reaches the required temperature. All angles are measured in radians.

b. Calculate the temperature of the heater 1 minute after it has reached the required temperature. (2 marks)

c. Find the difference between the maximum and minimum temperatures of the heater after it has reached the required temperature. (2 marks)

d. Calculate the times during the first 8 minutes, after the heater has reached the required temperature, when it reaches maximum temperature. (4 marks)

7. E/P

a. Express $132\sin x - 72\cos x$ in the form $R\sin(x-\alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. Give α to 1 decimal place. (4 marks)

The share price of a company during the first year after it is floated on the stock exchange can be modelled by the equation

$$P = 200 + 132\sin(0.8t) - 72\cos(0.8t)$$

where P is the share price in pence and t is the time in months since the company was floated (assuming it was floated on the 1st of January). All terms in t are measured in radians.

b. Write down the maximum share price and the month in which this first occurred. (3 marks)

c. Write down the month in which the shares were at their lowest price, and find the lowest share price. (3 marks)

d. State during which month the share price fell below 100p. (2 marks)

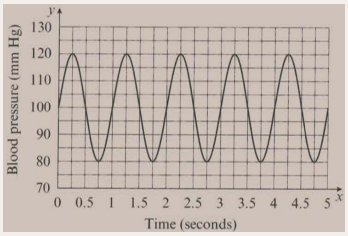
Luke purchased 2000 shares in the company at its flotation price, and sold them at the end of 12 months.

e. Find the profit or loss made by Luke. Give your answer to 2 significant figures. (2 marks)

Problem solving: Set A

Bronze

The graph shows a curve that models the blood pressure of a patient during the first 5 seconds of measurement.



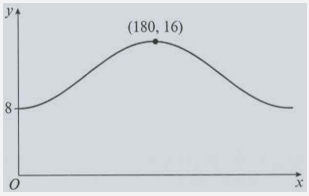
The model has the equation
$y = a + b\sin(cx)^\circ$
where y is blood pressure in mm Hg and x is time in seconds, and a, b and c are real constants.

a. By considering the maximum and minimum values of y, find the values of a and b. (3 marks)

b. By considering the period of the curve, find the value of c, and hence write the equation of the curve. (2 marks)

Silver

The graph shows a curve that models daylight hours on any day of the year in a particular city.



The model has the equation
$y = a + b\sin(x - c)^\circ$
where y is the number of daylight hours, and x is the number of days after 1 January, and a, b and c are real constants.
There are a minimum of 8 daylight hours at the start of the year, rising to a maximum of 16 daylight hours on day 180.

a. Find a, b and c and write the equation of the model. (4 marks)

b. Describe two transformations that could be made to the curve with equation $y = \sin x$ to obtain the curve given in the model. (2 marks)

c. Find the number of days in the year which experience more than 14 hours of sunshine. (2 marks)

Gold

The height of a car on a big wheel is modelled by the equation
$h = a + b\cos(ct - d)^\circ$
where h is the height of the car above ground in metres, t is time after the ride starts, in seconds, and a, b, c and d are real constants.
The initial height of the car is 2 m, and the maximum height reached on the ride is 22 m, which occurs 90 seconds after the start of the ride. The car completes four full revolutions, and the ride ends with the car in its initial position after 12 minutes.

a. Find the values of a, b, c and d and hence write the equation of the curve. (5 marks)

The fairground operator decides to increase the speed of the wheel so that it completes five full revolutions in 12 minutes.

b. Amend the model to reflect this change. (2 marks)

Revathi says that the length of time the car is more than 20 m above the ground will be the same for both models.

c. Determine whether Revathi is correct. (2 marks)

Problem solving: Set B

Bronze

a. Express $12\sin 2x - 5\cos 2x$ in the form $R\sin(2x - \alpha)$, with $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.
Give the value of α in radians to 3 decimal places. (4 marks)

b. Show that $12 - 3\operatorname{cosec} 2x = 5\cot 2x$ can be written as $12\sin 2x - 5\cos 2x = k$, where k is a positive constant to be determined. (2 marks)

c. Hence solve $12 - 3\operatorname{cosec} 2x = 5\cot 2x$ in the interval $0 \leq x \leq 2\pi$ giving your answers to 1 decimal place. (4 marks)

Silver

a. Express $7\cos^2 x - 3\sin^2 x - 8\sin x\cos x$ in the form $a\cos 2x - b\sin 2x + c$, where a, b and c are constants to be found. (3 marks)

b. Solve the equation $7\cos^2 x - 3\sin^2 x - 8\sin x\cos x = 4$ in the interval $0 \leq x \leq 2\pi$. Give your answers to 2 decimal places. (4 marks)

c.

i. Show that $10\cos^2 x - 8\sin x\cos x$ can be written in the form $a\cos 2x - b\sin 2x + k$, stating the values of a, b, and k.

ii. Hence deduce the exact maximum value of $10\cos^2 x - 8\sin x\cos x$.

(4 marks)

Gold

a. Express $\frac{15}{2}\sin 4x - 4\cos 4x - 4\cos 2x - 4$ in the form $\cos 2x(a\sin 2x - b\cos 2x - c)$, where a, b and c are constants to be found. (3 marks)

Hence or otherwise:

b. solve $\frac{15}{2}\sin 4x = 4(\cos 4x + \cos 2x + 1)$ in the interval $0 \leq x \leq \pi$, giving your answers to 2 decimal places (4 marks)

c. deduce the maximum value of $\dfrac{1}{30\sin x\cos x - 16\cos^2 x}$ (4 marks)

Now try this ⟶ Exam question bank Q11, Q17, Q36, Q40, Q50, Q59, Q71, Q83

Selected exam question bank questions

Question 11

a. Use the expansion of $\sin(2x + x)$ to write $\sin 3x$ in terms of $\sin x$. (3 marks)

b. Hence solve the equation $\sin 3x = \sin x$ in the interval $0 \leq x \leq \pi$. (3 marks)

Question 17

Prove that $\dfrac{1 + \tan^2 x}{1 - \tan^2 x} \equiv \sec 2x$ (4 marks)

Question 36

The curve C has parametric equations

$$x = \cos t, \quad y = \cos\left(t + \frac{\pi}{3}\right), \quad 0 < t < \pi$$

Show that a Cartesian equation of the curve is $y = \dfrac{1}{2}x - \dfrac{\sqrt{3}}{2}\sqrt{1 - x^2}$, for $a < x < b$, stating the values of a and b. (5 marks)

Question 40

a. Use $\tan(x + \alpha) = \dfrac{\tan x + \tan \alpha}{1 - \tan x \tan \alpha}$ to show that $\tan 105° = -2 - \sqrt{3}$. (4 marks)

b. Hence or otherwise, show that $\cot 105° = -2 + \sqrt{3}$. (3 marks)

Question 50

a. Prove that $\sin 2\theta - \tan \theta \equiv \tan \theta \cos 2\theta$. (3 marks)

b. Hence solve the following equation in the interval $-\dfrac{\pi}{2} < x < \dfrac{\pi}{2}$

$$(\sec^2 x + 3)(\sin 2x - \tan x) = 4 \tan^2 x \cos 2x$$

Give your answers to 3 decimal places where necessary. (6 marks)

Question 59

In triangle ABC, $AB = 3\,\text{cm}$, $BC = 7\,\text{cm}$ and $AC = 8\,\text{cm}$ and $\angle ACB = \dfrac{\theta}{2}$

a. Use the cosine rule to show that $\cos \dfrac{\theta}{2} = \dfrac{13}{14}$. (3 marks)

b. Hence find the exact value of $\cos 2\theta$. (3 marks)

Question 71

a. Express $\dfrac{3}{2}\cos\theta + \dfrac{5}{2}\sin\theta$ in the form $R\cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$.
Give the value of α to two decimal places. (3 marks)

b. Hence find the maximum value of $\dfrac{16}{5} + \left(\dfrac{3}{2}\cos\theta + \dfrac{5}{2}\sin\theta\right)^2$. (2 marks)

c. Solve, for $-\pi < \theta < \dfrac{\pi}{2}$, the equation $\dfrac{3}{2}\cos 2\theta + \dfrac{5}{2}\sin 2\theta = 2$, giving your answers to two decimal places. (5 marks)

Question 83

a. Express $12.5\cos x + 6.5\sin x$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \dfrac{\pi}{2}$.
Give the value of α to 1 decimal place. (4 marks)

b.

i. Find the maximum value of $12.5\cos x + 6.5\sin x$.

ii. Find the value of x, for $0 < x < \pi$, at which the maximum occurs.

(3 marks)

The height above ground of a particular car on a Ferris wheel can be modelled by the function

$$H = 16 - 12.5\cos(1.05t) - 6.5\sin(1.05t)$$

where H is the height of the car above ground in metres, and t is time in minutes after the ride begins. All angles are measured in radians. Give your answers to 1 decimal place where necessary.

c. State the maximum height of the wheel and the time at which this maximum occurs. (3 marks)

d. Calculate the height of the car at 4 minutes. (2 marks)

e. Calculate the times the car reaches a height of 15 metres during the first 6 minutes. (4 marks)

f. Find the time it takes for the Ferris wheel to complete one rotation. (2 marks)

Answers

7.1 Addition formulae

1.

a. $\sin((90° - A) + B) = \sin(90° - A)\cos B + \cos(90° - A)\sin B$
$\sin(90° - (A - B)) = \sin(90° - A)\cos B + \cos(90° - A)\sin B$
$\sin(90° - A) = \cos A$ and $\cos(90° - A) = \sin A$, therefore
$\cos(A - B) = \cos A \cos B + \sin A \sin B$

b. $\cos(A - (-B)) = \cos A \cos(-B) + \sin A \sin(-B)$
$\cos(-B) = \cos B$ and $\sin(-B) = -\sin B$, therefore
$\cos(A + B) = \cos A \cos B - \sin A \sin B$

2.

a. $\dfrac{1}{2}\sin x + \dfrac{\sqrt{3}}{2}\cos x$

b. $\dfrac{\sqrt{3}}{2}\cos x + \dfrac{1}{2}\sin x$

c. $\dfrac{\tan x + \sqrt{3}}{1 - \sqrt{3}\tan x}$

3.

a. $\tan 4\theta$

b. $\cos 2a$

c. $\sin(x - 45°)$

4. $\dfrac{1 - \tan x \tan y}{\tan x + \tan y}$

5.

a. $\cos\left(x + \dfrac{\pi}{3}\right)$

b. $\sin\left(x - \dfrac{\pi}{6}\right)$

6. $\sin\left(x + \dfrac{\pi}{6}\right) = \cos x \Rightarrow \sin x \cos\dfrac{\pi}{6} + \cos x \sin\dfrac{\pi}{6} = \cos x$
$\Rightarrow \sqrt{3}\sin x + \cos x = 2\cos x \Rightarrow \sqrt{3}\sin x = \cos x$
$\Rightarrow \tan x = \dfrac{1}{\sqrt{3}}$

7. $2\sin\left(x + \dfrac{5\pi}{6}\right) + 2\sin\left(x + \dfrac{10\pi}{6}\right)$
$= 2\left(-\dfrac{\sqrt{3}}{2}\sin x + \dfrac{1}{2}\cos x\right) + 2\left(\dfrac{1}{2}\sin x - \dfrac{\sqrt{3}}{2}\cos x\right)$
$= (1 - \sqrt{3})\sin x + (1 - \sqrt{3})\cos x$

7.2 Using the angle addition formulae

1.

a. $\dfrac{\sqrt{3}}{2}$

b. $\dfrac{\sqrt{2}}{2}$

c. $\dfrac{\sqrt{3}}{3}$

2.

a. $\sin 135° = \sin(90° + 45°)$
$= \sin 90° \cos 45° + \cos 90° \sin 45°$

$= (1)\left(\frac{\sqrt{2}}{2}\right) + (0)\left(\frac{\sqrt{2}}{2}\right) = \frac{\sqrt{2}}{2}$

b. $\cos 15° = \cos(45° - 30°)$
$= \cos 45° \cos 30° + \sin 45° \sin 30°$
$= \left(\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right) = \frac{\sqrt{6}+\sqrt{2}}{4}$

c. $\tan 75° = \tan(45° + 30°)$
$= \frac{\tan 45° + \tan 30°}{1 - \tan 45° \tan 30°}$
$= \frac{\sqrt{3}+1}{\sqrt{3}-1}$
$= \frac{(\sqrt{3}+1)(\sqrt{3}+1)}{(\sqrt{3}-1)(\sqrt{3}+1)}$
$= \frac{4+2\sqrt{3}}{2}$
$= 2 + \sqrt{3}$

3.

a. $\frac{13}{85}$

b. $\frac{36}{77}$

c. $\frac{85}{84}$

4.

a. $\frac{\sqrt{3}}{2}$

b. $-\sqrt{3}$

c. $\frac{1}{2}$

5.

a. $-\frac{36}{325}$

b. $\frac{204}{253}$

c. $\frac{325}{323}$

6.

a. $\frac{\tan\frac{\pi}{4} - \tan\frac{\pi}{6}}{1 + \tan\frac{\pi}{4}\tan\frac{\pi}{6}}$

b. $\tan\frac{\pi}{12} = \frac{1 - \frac{\sqrt{3}}{3}}{1 + \left(1 \times \frac{\sqrt{3}}{3}\right)} = 2 - \sqrt{3}$

7.

a. $\sin 105° = \sin 60° \cos 45° + \cos 60° \sin 45°$
$= \frac{\sqrt{3}}{2} \times \frac{\sqrt{2}}{2} + \frac{1}{2} \times \frac{\sqrt{2}}{2} = \frac{\sqrt{6}+\sqrt{2}}{4}$

b. $\operatorname{cosec} 105° = \frac{4}{\sqrt{6}+\sqrt{2}} = \frac{4}{\sqrt{6}+\sqrt{2}} \times \frac{\sqrt{6}-\sqrt{2}}{\sqrt{6}-\sqrt{2}} = \sqrt{6} - \sqrt{2}$

7.3 Double-angle formulae

1.

a. $\sin\frac{2\pi}{3}$

b. $2\cos 14°$

c. $\cot 27°$

2.

a. $3\sin 10\theta$

b. $\sin 2\theta$

c. $\cot\theta$

3.

a. $-\dfrac{120}{119}$

b. $\dfrac{119}{169}$

c. $-\dfrac{169}{120}$

4.

a. $y = \dfrac{4}{5}x^2 - 8$

b. $y = \sqrt[4]{\dfrac{2}{x+1}}$

5.

a. $\cos^4 x - \sin^4 x \equiv (\cos^2 x + \sin^2 x)(\cos^2 x - \sin^2 x)$
$\equiv \cos 2x$

b. $\dfrac{\sqrt{3}}{2}$

6.

a. $c = 1 - \dfrac{a^2}{2}$

b. $\sin x = \dfrac{a}{2}$ and $\cos x = \dfrac{b}{2}$, so $\tan x = \dfrac{a}{b}$
$d = \tan 2x = \dfrac{2\tan x}{1 - \tan^2 x}$
$= \dfrac{2\left(\frac{a}{b}\right)}{1 - \left(\frac{a}{b}\right)^2}$
$= \dfrac{2ab}{b^2 - a^2}$

7. $\dfrac{\sqrt{2}}{10}$

7.4 Solving trigonometric equations

1.

a. $x = 3.0, 6.1$

b. $x = 0.5, 3.7$

c. $x = 1.2, 5.1$

2.

a. $\theta = 63.4°, 116.6°$

b. $\theta = 0, 180°, 82.8°$

c. $\theta = 37.8°, 142.2°$

3.

a. $\cos\left(x - \dfrac{\pi}{6}\right) = \cos x \cos\dfrac{\pi}{6} + \sin x \sin\dfrac{\pi}{6}$
$\equiv \dfrac{\sqrt{3}}{2}\cos x + \dfrac{1}{2}\sin x \equiv \dfrac{1}{2}(\sqrt{3}\cos x + \sin x)$

b. $x = \dfrac{\pi}{2}, \dfrac{11\pi}{6}$

4.

a. $\sin(x - 45°) = \cos x \Rightarrow \frac{\sqrt{2}}{2}\sin x - \frac{\sqrt{2}}{2}\cos x = \cos x$
$\Rightarrow \tan x = 1 + \sqrt{2}$

b. $x = 67.5°, 247.5°$

5.

a. $2\sin\left(x - \frac{\pi}{6}\right) = \sin\left(x + \frac{\pi}{2}\right)$
$\Rightarrow 2\sin x\cos\frac{\pi}{6} - 2\cos x\sin\frac{\pi}{6} = \sin x\cos\frac{\pi}{2} + \cos x\sin\frac{\pi}{2}$
$\Rightarrow \sqrt{3}\sin x - \cos x = \cos x \Rightarrow \sqrt{3}\sin x - 2\cos x = 0$

b. $x = -2.28, 0.86$

6.

a. $\tan 2x - \tan x = 0 \Rightarrow \frac{2\tan x}{1 - \tan^2 x} - \tan x = 0$
$\Rightarrow 2\tan x - \tan x + \tan^3 x = 0 \Rightarrow \tan x + \tan^3 x = 0$

b. $x = 0, \pi, 2\pi$

7.

a. $4\cos^2 x + \cos x - 3 = 0$

b. $x = -\pi, -0.72, 0.72, \pi$

8. $x = -\pi, -0.59, 0, 2.55$

7.5 Simplifying $a\cos x \pm b\sin x$

1.

a. $R = 29$, $\tan\alpha = \frac{21}{20}$

b. $R = \sqrt{8} = 2\sqrt{2}$, $\tan\alpha = \frac{\sqrt{5}}{\sqrt{3}}$

c. $R = \sqrt{13}$, $\tan\alpha = \frac{3}{2}$

2.

a. $R = \sqrt{5}$, $\alpha = 63.4°$

b. $R = 13$, $\alpha = 22.6°$

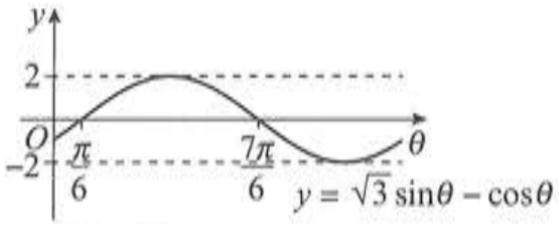
c. $R = 2\sqrt{5}$, $\alpha = 26.6°$

3.

a.

i. $2\sin\left(\theta - \frac{\pi}{6}\right)$

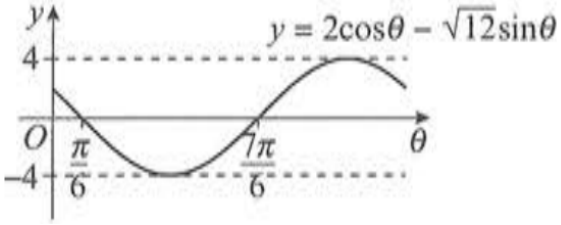
ii.



b.

i. $4\cos\left(\theta + \frac{\pi}{3}\right)$

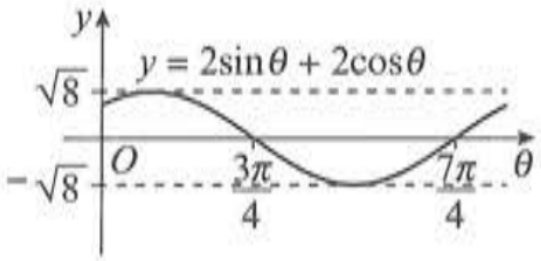
ii.



c.

i. $\sqrt{8}\sin\left(\theta + \frac{\pi}{4}\right)$

ii.



4.

a. $R = \sqrt{20} = 2\sqrt{5}$, $\alpha = 63.43°$

b. $\theta = 74.4°, 338.7°$

5.

a. $\sqrt{13}\cos(2\theta - 0.588)$

b. $\theta = -2.20, -0.35, 0.94, 2.79$

c. $k < -\sqrt{13}$, $k > \sqrt{13}$

6.

a. $17\sin(\theta - 1.08)$

b.

i. $\frac{5}{3}$

ii. $\theta = 5.79$

c.

i. $\frac{1}{4}$

ii. $\theta = 2.65$

7.

a. $2\sin\left(x - \frac{\pi}{6}\right) - (\sqrt{3} - 2)\sin x$

$= 2\left(\frac{\sqrt{3}}{2}\sin x - \frac{1}{2}\cos x\right) - \sqrt{3}\sin x + 2\sin x$

$= 2\sin x - \cos x$

b. $x = -2.90, 0.69, 3.38$

c. 25

7.6 Proving trigonometric identities

1.

a. $\frac{\sin 2x - \tan x}{\tan x} \equiv \frac{2\sin x\cos x}{\frac{\sin x}{\cos x}} - 1 \equiv 2\cos^2 x - 1 \equiv \cos 2x$

b. $\dfrac{\sin 6\theta}{1-\cos 6\theta} \equiv \dfrac{2\sin 3\theta\cos 3\theta}{1-(1-2\sin^2 3\theta)}$
$\equiv \dfrac{2\sin 3\theta\cos 3\theta}{2\sin^2 3\theta} \equiv \dfrac{\cos 3\theta}{\sin 3\theta} \equiv \cot 3\theta$

c. $\dfrac{1}{4}\sec x\tan 2x \equiv \sec x\dfrac{2\tan x}{4(1-\tan^2 x)} = \dfrac{\sec x\tan x}{2(1-\tan^2 x)}$
$\equiv \dfrac{\frac{1}{\cos x}\times\frac{\sin x}{\cos x}}{2\left(1-\frac{\sin^2 x}{\cos^2 x}\right)} = \dfrac{\frac{\sin x}{\cos^2 x}}{2\left(\frac{\cos^2 x-\sin^2 x}{\cos^2 x}\right)}$
$\equiv \dfrac{\sin x}{2(\cos^2 x-\sin^2 x)} \equiv \dfrac{1}{2}\sec 2x\sin x$

2.

a. $\cos(x-30^\circ)-\cos(x+30^\circ)$
$\equiv (\cos x\cos 30^\circ+\sin x\sin 30^\circ)-(\cos x\cos 30^\circ-\sin x\sin 30^\circ)$
$\equiv \left(\dfrac{\sqrt{3}}{2}\cos x+\dfrac{1}{2}\sin x\right)-\left(\dfrac{\sqrt{3}}{2}\cos x-\dfrac{1}{2}\sin x\right) \equiv \sin x$

b. $\dfrac{\sin(a+b)}{\sin(a-b)} \equiv \dfrac{\sin a\cos b+\cos a\sin b}{\sin a\cos b-\cos a\sin b}$
$\equiv \dfrac{\frac{\sin a\cos b}{\sin a\sin b}+\frac{\cos a\sin b}{\sin a\sin b}}{\frac{\sin a\cos b}{\sin a\sin b}-\frac{\cos a\sin b}{\sin a\sin b}} \equiv \dfrac{\cot b+\cot a}{\cot b-\cot a}$

c. $\cot\left(\dfrac{\pi}{4}-\theta\right) \equiv \dfrac{1+\tan\frac{\pi}{4}\tan\theta}{\tan\frac{\pi}{4}-\tan\theta} \equiv \dfrac{1+\tan\theta}{1-\tan\theta}$
$\equiv \dfrac{1+\frac{\sin\theta}{\cos\theta}}{1-\frac{\sin\theta}{\cos\theta}} \equiv \dfrac{\frac{\cos\theta+\sin\theta}{\cos\theta}}{\frac{\cos\theta-\sin\theta}{\cos\theta}} \equiv \dfrac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}$

3.

a. $2\sin\left(3\theta+\dfrac{\pi}{3}\right) \equiv 2\sin 3\theta\cos\dfrac{\pi}{3}+2\cos 3\theta\sin\dfrac{\pi}{3}$
$\equiv \left(2\times\dfrac{1}{2}\right)\sin 3\theta+\left(2\times\dfrac{\sqrt{3}}{2}\right)\cos 3\theta$
$\equiv \sin 3\theta+\sqrt{3}\cos 3\theta$

b. $\cos 3A \equiv \cos(2A+A) \equiv \cos 2A\cos A-\sin 2A\sin A$
$\equiv (2\cos^2 A-1)\cos A-(2\sin A\cos A)\sin A$
$\equiv 2\cos^3 A-\cos A-2\sin^2 A\cos A$
$\equiv 2\cos^3 A-\cos A-2(1-\cos^2 A)\cos A$
$\equiv 2\cos^3 A-\cos A-2\cos A+2\cos^3 A$
$\equiv 4\cos^3 A-3\cos A$

c. $\cot 2x+\tan x \equiv \dfrac{\cos 2x}{\sin 2x}+\dfrac{\sin x}{\cos x}$
$\equiv \dfrac{\cos 2x\cos x+\sin x\sin 2x}{\sin 2x\cos x}$
$\equiv \dfrac{\cos(2x-x)}{\sin 2x\cos x} \equiv \dfrac{\cos x}{\sin 2x\cos x}$
$\equiv \operatorname{cosec} 2x$

4. $\sqrt{3}\cos x-\sin x \equiv 2\cos 30^\circ\cos x-2\sin 30^\circ\sin x$
$\equiv 2\cos(x+30^\circ)$

5.

a. $\dfrac{\sin 2x}{1+\cos 2x} \equiv \dfrac{2\sin x\cos x}{1+(2\cos^2 x-1)} \equiv \dfrac{\sin x}{\cos x} \equiv \tan x$

b. $\sqrt{3}$

6. $2\sin\left(2\theta-\dfrac{\pi}{4}\right) \equiv 2\sin 2\theta\cos\dfrac{\pi}{4}-2\cos 2\theta\sin\dfrac{\pi}{4}$
$\equiv \sqrt{2}\sin 2\theta-\sqrt{2}\cos 2\theta$
$\equiv \sqrt{2}(2\sin\theta\cos\theta)-\sqrt{2}(1-2\sin^2\theta)$

$\equiv 2\sqrt{2}\sin\theta\cos\theta - \sqrt{2} + 2\sqrt{2}\sin^2\theta$

7.

a. $\cos^4 x - \sin^4 x \equiv (\cos^2 x + \sin^2 x)(\cos^2 x - \sin^2 x)$
$\equiv 1 \times \cos 2x \equiv \cos 2x$

b. $\frac{1}{2}$

7.7 Modelling with trigonometric functions

1.

a. 0.6 m

b. 0.75 seconds

c. 20

d. A body moving in the air is better modelled by a parabola.

2.

a. 40.5 cm

b. 43 cm

c. 0.5, 2.5, 4.5 seconds

d. Minimum depth of water is 38 cm, so pond is not deep enough.

3.

a. 130 mm Hg

b. 80 mm Hg

c. 105 mm Hg

d. 0.03, 0.37, 0.83, 1.17, 1.63, 1.97 seconds

4.

a. 9000 in April

b. March, April and May

c. 4760 toys

d. This is extrapolation, so he may not be correct.

5.

a. $5\cos(\theta + 45°)$

b.

i. -5

ii. 135°

c. 14 m at 04:30

d. 4 m at 22:30

e.

i. 01:30

ii. 07:30

6.

a. $\frac{\sqrt{65}}{20}\cos(x - 1.1)$

b. 30.25°C

c. 0.8°C

d. 0.6 minutes, 3.7 minutes, 6.8 minutes

7.

a. $12\sqrt{157}\sin(x - 0.5)$

b. 350p in March

c. 50p during July
d. June
e. £2400 profit

Problem solving: Set A

Bronze

a. $a = 100, b = 20$
b. $c = 360$; $y = 100 + 20\sin(360x)^\circ$

Silver

a. $y = 12 + 4\sin(x - 90^\circ)$
b. Vertical stretch with scale factor 4, translation by $\begin{pmatrix} 90 \\ 12 \end{pmatrix}$
c. 120 days

Gold

a. $y = 12 + 10\cos(2t - 180^\circ)$
b. $y = 12 + 10\cos\left(\frac{5}{2}t - 180^\circ\right)$
c. Revathi is correct.

Problem solving: Set B

Bronze

a. $13\sin(2x - 0.395)$
b. $12 - 3\operatorname{cosec} 2x = 5\cot 2x \Rightarrow 12 - \dfrac{3}{\sin 2x} = \dfrac{5\cos 2x}{\sin 2x}$
$\Rightarrow 12\sin 2x - 5\cos 2x = 3$
c. $x = 0.3, 1.7, 3.5, 4.8$

Silver

a. $5\cos 2x - 4\sin 2x + 2$
b. $x = 0.29, 2.18, 3.43, 5.32$
c.
i. $10\cos^2 x - 8\sin x\cos x$
$\equiv 10\left(\dfrac{1 + \cos 2x}{2}\right) - 4(2\sin x\cos x)$
$= 5\cos 2x - 4\sin 2x + 5$
ii. $5 + \sqrt{41}$

Gold

a. $\cos 2x(15\sin 2x - 8\cos 2x - 4)$
b. $x = 0.36, 0.79, 1.70, 2.36$
c. $\dfrac{1}{9}$

Selected exam question bank answers

Question 11

a. $\sin 3x = 3\sin x - 4\sin^3 x$

b. $x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$

Question 17 $\frac{1+\tan^2 x}{1-\tan^2 x} \equiv \frac{1+\frac{\sin^2 x}{\cos^2 x}}{1-\frac{\sin^2 x}{\cos^2 x}} \equiv \frac{\cos^2 x+\sin^2 x}{\cos^2 x-\sin^2 x}$

$\equiv \frac{\cos^2 x+\sin^2 x}{\cos^2 x-\sin^2 x} \equiv \frac{1}{\cos 2x} \equiv \sec 2x$

Question 36 $y = \cos\left(t+\frac{\pi}{3}\right) = \cos t\cos\frac{\pi}{3} - \sin t\sin\frac{\pi}{3}$

$= \frac{1}{2}\cos t - \frac{\sqrt{3}}{2}\sin t$

$= \frac{1}{2}x - \frac{\sqrt{3}}{2}\sqrt{1-x^2},\ -1 < x < 1$

Question 40

a. $\tan(60^\circ+45^\circ) = \frac{\tan 60^\circ + \tan 45^\circ}{1-\tan 60^\circ \tan 45^\circ}$

$= \frac{\sqrt{3}+1}{1-(\sqrt{3}\times 1)} = -2-\sqrt{3}$

b. $\cot 105^\circ = \frac{1}{-2-\sqrt{3}} \Rightarrow \frac{1}{-2-\sqrt{3}} \times \frac{-2+\sqrt{3}}{-2+\sqrt{3}} = -2+\sqrt{3}$

Question 50

a. $\sin 2\theta - \tan\theta \equiv 2\sin\theta\cos\theta - \frac{\sin\theta}{\cos\theta}$

$\equiv \frac{2\sin\theta\cos^2\theta - \sin\theta}{\cos\theta}$

$\equiv \frac{\sin\theta(2\cos^2\theta - 1)}{\cos\theta}$

$\equiv \tan\theta\cos 2\theta$

b. $x = -\frac{\pi}{4}, 0, \frac{\pi}{4}, 1.107$

Question 59

a. $\cos\frac{\theta}{2} = \frac{8^2+7^2-3^2}{2\times 8\times 7} = \frac{13}{14}$

b. $\frac{239}{4802}$

Question 71

a. $\frac{\sqrt{34}}{2}\cos(\theta - 1.03)$

b. 11.7

c. $\theta = -3.03, -2.22, 0.11, 0.92$

Question 83

a. $\frac{\sqrt{794}}{2}\cos(x - 0.5)$

b.

i. 14.1 (3 s.f.)

ii. 0.5

c. 30.1 m, 3.5 minutes

d. 27.8 m (using original equation) or 28.0 m (using answer from a)

e. 1.9 minutes, 5.0 minutes

f. 6.0 minutes (2 s.f.)